

5.1.4 Ideal gases

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) amount of substance in moles; Avogadro constant N_A equals $6.02 \times 10^{23} \text{ mol}^{-1}$	assumptions for the model: large number of molecules in random, rapid motion particles (atoms or molecules) occupy negligible volume compared to the volume of gas all collisions are perfectly elastic and the time of the collisions is negligible compared to the time between collisions negligible forces between particles except during collision HSW1
(b) model of kinetic theory of gases	
(c) pressure in terms of this model	
(d) (i) the equation of state of an ideal gas $pV = nRT$, where n is the number of moles	HSW1, 2 Explanation of pressure in terms of Newtonian theory.
(ii) techniques and procedures used to investigate $pV = \text{constant}$ (Boyle's law) and $\frac{p}{T} = \text{constant}$	
(iii) an estimation of absolute zero using variation of gas temperature with pressure	
(e) the equation $pV = \frac{1}{3}Nmc^2$, where N is the number of particles (atoms or molecules) and c^2 is the mean square speed	Derivation of this equation is not required. HSW2
(f) root mean square (r.m.s.) speed; mean square speed	Learners should know about the general characteristics of the Maxwell-Boltzmann distribution.
(g) the Boltzmann constant; $k = \frac{R}{N_A}$	
(h) $pV = NkT$; $\frac{1}{2}mc^2 = \frac{3}{2}kT$	Learners will also be expected to know the derivation of the equation $\frac{1}{2}mc^2 = \frac{3}{2}kT$ from $pV = \frac{1}{3}Nmc^2$ and $pV = NkT$. HSW2
(i) internal energy of an ideal gas.	